Rayane El Mselmi

Durham • +44 7442 923586 • rayane.elmselmi@gmail.com • github.com/icyy001 • https://rayane-elmselmi.vercel.app

PROFILE

Second-year BSc Computer Science student at Durham University (Expected 2027) open to software, full-stack, backend, data and cybersecurity internships. Strong foundations in algorithms and systems, with practical experience building end-to-end web projects and communicating results through data-driven analysis.

• API design & implementation • Automated testing • Data benchmarking & visualisation • Clear documentation
• Fast learner • Calm under time pressure • Strong written communication

EDUCATION

Durham University — BSc Computer Science *(Expected 2027)*

- Year 1: 2:1 (aiming First Class / 2:1 in Year 2)
- Modules: Algorithms & Data Structures, Networks & Systems, Software Engineering, Data Science, AI

Durham University International Study Centre — Computer Science Foundation Year *(82% overall)*

PROJECTS

EventHub Pro — Events REST API (Express, Prisma/SQLite) github.com/icyy001/EventHub-Pro

- Implemented paginated, case-insensitive search: GET /events?search=&page=&limit= → { page, limit, total, totalPages, data }
- Built endpoints (POST /events, GET /events/:id, DELETE /events/:id) with 404/204 handling, seeded 25 Durham events
- Added helmet + CORS and a minimal JS/CSS homepage to list/search events end-to-end

EventHub — Web App + Tiny JSON API (Node/Express, Bootstrap) • github.com/icyy001/EventHub

- Built a Bootstrap/vanilla JS frontend connected to a Node/Express API for events and places
- Implemented 6 endpoints under /api (list, get-by-id, create), enabled CORS for local development
- Added Jest/Supertest tests (npm test) to verify API behaviour

Maze Generation Algorithms — Benchmarking Study (Python, NumPy, Matplotlib) • 76/100

- Benchmarked DFS, Prim's and Kruskal's across 50×50–200×200 grids over 100 iterations; compared time and memory
- Stored results as CSV datasets; visualised comparisons with Matplotlib charts/tables
- Produced a PDF research poster (methodology, literature review, conclusions) to communicate findings

Dynamic Event Website — JSON to DOM Rendering (Vanilla JS) • icyy001.github.io/Dynamic-Event-Website

- Loaded event data from a local JSON file via Fetch API (AJAX) and updated the DOM without reloads
- Implemented a minimal, responsive layout focused on clarity and usability

SKILLS

Languages: Python, Java, JavaScript, HTML/CSS; **Backend/DB:** Node.js, Express, REST APIs, Prisma, SQLite;
Testing/Data/Tools: Jest, Supertest, Git, Bootstrap; NumPy, Pandas, Matplotlib

ACTIVITIES & LEADERSHIP

Cybersecurity CTF Assessment (University) — 80+/100 (VM-based).

Class Representative (High School) — 2 years.

LANGUAGES

Arabic (Fluent) • French (C1) • English (C1) • Spanish (A1–A2)